TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
Semester II 2026

Simplex Method Project

Dr. Francisco J. Torres-Rojas

Dayron Josué Porras-Pérez
2023124693

Javier Lee-Liang
2020184824

José Julián Araya-Castillo
2020098909

San Jose, Costa Rica
November 2026

The Simplex Algorithm

The Simplex Algorithm is a mathematical method used to solve linear programming problems. It was developed by George Dantzig in 1947 and has since become one of the most widely used optimization techniques in various fields, including operations research, economics, and engineering.

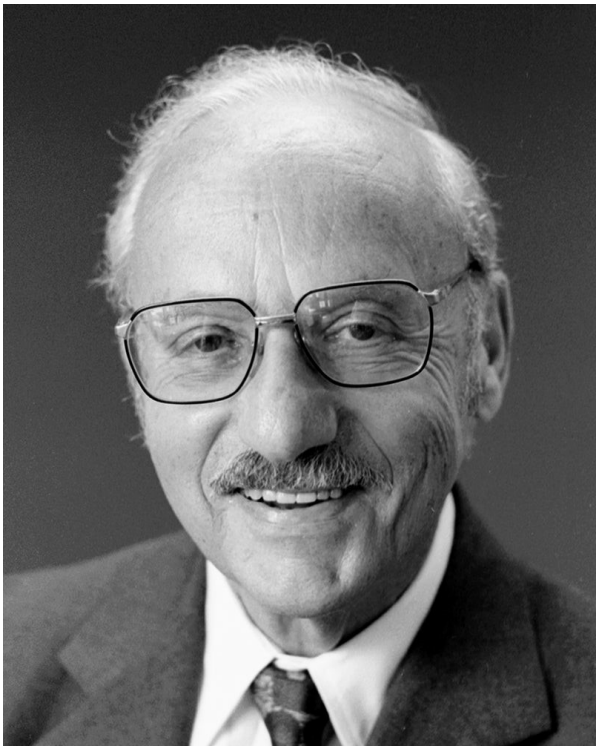


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

Original Problem

Description

- Problem Name:
- Amount of decision variables: 15
- Amount of restrictions: 2
- Total Amount of variables (decision + slack): 17
- Type of problem: Maximization

Equations

$$\text{Max } Z = 3\,x_1 + 4\,x_2 + 4\,x_3 + 3\,x_4 + 4\,x_5 + 3\,x_6 + 3\,x_7 + 3\,x_8 + 4\,x_9 - 5\,x_{10} + 3\,x_{11} + 8\,x_{12} - 2\,x_{13} + 3\,x_{14} + 6\,x_{15}$$

subject to:

$$3\,x_1 + 4\,x_5 + 3\,x_8 \leq 0$$

$$6\,x_2 + 6\,x_3 + 2\,x_6 + 3\,x_7 \leq 0$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{11}, x_{12}, x_{13}, x_{14}, x_{15} \geq 0$$

Initial Table

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	s_1	s_2	b
1.0000	-3.0000	-4.0000	-4.0000	-3.0000	-4.0000	-3.0000	-3.0000	-3.0000	-4.0000	5.0000	-3.0000	-8.0000	2.0000	-3.0000	-6.0000	0.0000	0.0000	0.0000
0.0000	3.0000	0.0000	0.0000	0.0000	4.0000	0.0000	0.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000
0.0000	0.0000	6.0000	6.0000	0.0000	0.0000	2.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000

Intermediate Tables

Iteration 1

The entering variable would be x_{12} (coefficient -8.0000 in the Z row), but every entry of its column in the constraints is less than or equal to zero, so no row limits its growth: the problem is **unbounded**.

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	s_1	s_2	b
1.0000	-3.0000	-4.0000	-4.0000	-3.0000	-4.0000	-3.0000	-3.0000	-3.0000	-4.0000	5.0000	-3.0000	-8.0000	2.0000	-3.0000	-6.0000	0.0000	0.0000	0.0000
0.0000	3.0000	0.0000	0.0000	0.0000	4.0000	0.0000	0.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000
0.0000	0.0000	6.0000	6.0000	0.0000	0.0000	2.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000

Final Table

Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	s_1	s_2	b
1.0000	-3.0000	-4.0000	-4.0000	-3.0000	-4.0000	-3.0000	-3.0000	-3.0000	-4.0000	5.0000	-3.0000	-8.0000	2.0000	-3.0000	-6.0000	0.0000	0.0000	0.0000
0.0000	3.0000	0.0000	0.0000	0.0000	4.0000	0.0000	0.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000
0.0000	0.0000	6.0000	6.0000	0.0000	0.0000	2.0000	3.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000	0.0000

∞

Solution

Solution

The problem is **unbounded**: there is no finite optimal solution.

Unbounded problem

In iteration 1 the variable x_{12} has an improving coefficient in the Z row (-8.0000), but all the entries of its column in the constraint rows are ≤ 0. Therefore the ratio test has no candidate row and x_{12} can grow indefinitely without violating any constraint, improving Z without limit. The problem has no finite optimal solution.